

European Option Convergence Study

Neel Dev Sen

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Introduction

This convergence study is based on the 3 European Option pricers I have made in my other project called Pricing Engines (which you can find on my GitHub). The numerical methods that will be compared is the Binomial Tree, Trinomial Tree and the Finite Differences (using the Crank-Nicolson scheme)

1 Theoretical and Numerical Framework

This section will mainly just be stating the Time and Space complexities, the complete derivations of these are in this document.

1.1 Time Complexities

The time complexity of a Binomial Tree with N steps is:

$$\mathcal{O}(N^2) \tag{1}$$

The time complexity of a Trinomial Tree with N steps is also:

$$\mathcal{O}(N^2) \tag{2}$$

Finally the time complexity of the Finite Differences with a mesh of M space nodes and N time nodes is:

$$\mathcal{O}(M \times N) \tag{3}$$

Since I used the Thomas algorithm it makes it faster instead of $\mathcal{O}(M^3 \times N)$. In this experiment to keep this all similar so it is easier to compare, I define $M = N$ to make a square mesh such that the time complexity of Finite Differences is also:

$$\mathcal{O}(N^2) \tag{4}$$

All of these time complexities will be useful when I show the runtime-convergence graphs of these numerical methods

1.2 Space Complexities

The total number of bytes allocated in a Binomial Tree with N steps is: $(2N - 1) \times$ size of type. Hence the space complexity of the Binomial Tree is:

$$\mathcal{O}(N) \tag{5}$$

The total number of bytes allocated in a Trinomial Tree with N steps is: $(4N) \times$ size of type. Hence the space complexity of the Trinomial Tree is again:

$$\mathcal{O}(N) \tag{6}$$

Finally the space requirement of the Finite Differences with a mesh of N space nodes and N time nodes stores $(4N - 2) \times$ size of type bytes of memory. This gives us the space complexity of the Finite Differences to be:

$$\mathcal{O}(N) \tag{7}$$

1.3 Error Metrics

The absolute error is the main error metric that will be used to show convergence and it goes as shown.

$$\text{Absolute Error} = |V_N - V_{CF}| \quad (8)$$

where V_N is the value of the option given by the numerical method, and V_{CF} is the value of the option given by the closed form solution.

For the Arbitrage Error, I will just look at how far the options are from meeting the Put-Call Parity (the no-arbitrage condition). The Put Call Parity is as shown

$$C_t - P_t = S - Ke^{-r(T-t)} \quad (9)$$

The arbitrage error will be defined as:

$$\text{Arbitrage Error} = |C_t - P_t - S + Ke^{-r(T-t)}| \quad (10)$$

1.4 Data

The table below will be the data that is used in the convergence study.

Parameter	Value
Spot Price S_0	120.0
Strike K	100.0
Risk-Free Rate r	0.04
Volatility σ	0.25
Maturity T	2.0
Domain Multiplier	4.0

Table 1: Parameters used for the numerical convergence study.

2 Results

2.1 Black-Scholes Call

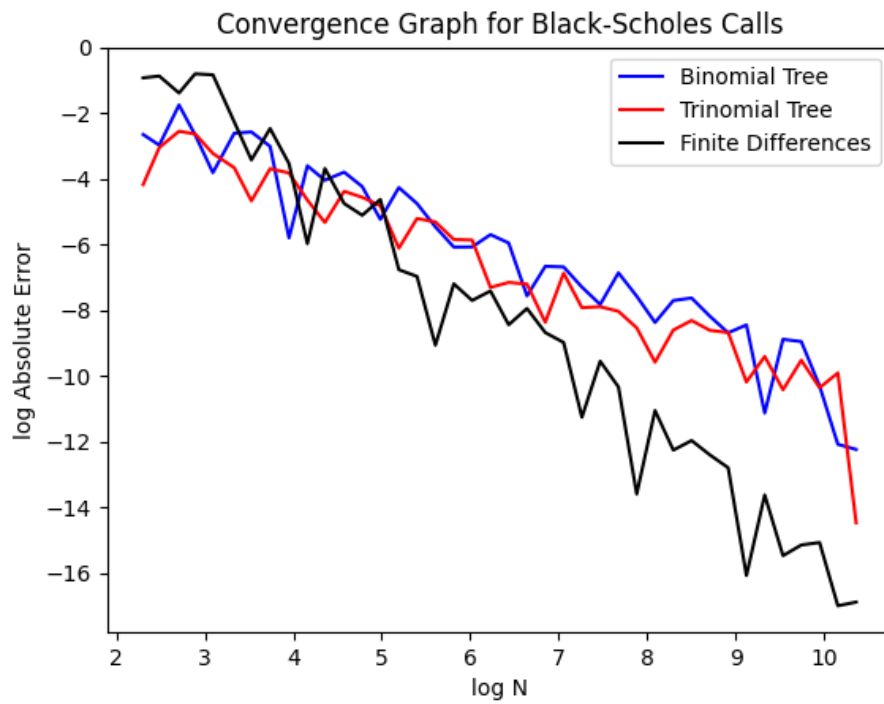


Figure 1: Convergence of the Black-Scholes call price.

2.2 Regression on the Convergence

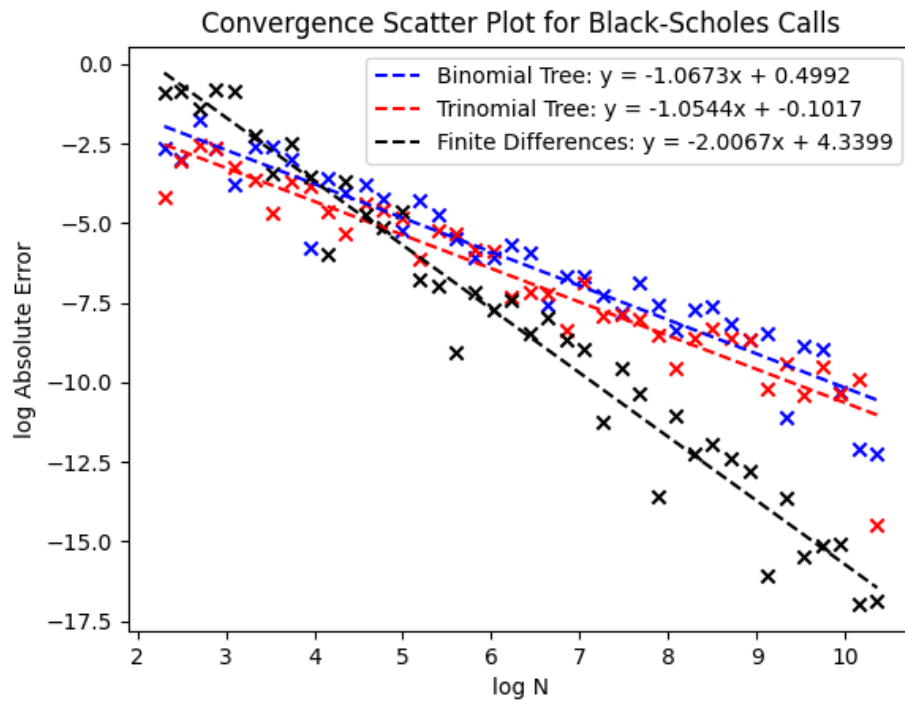


Figure 2: Convergence of the Black-Scholes call price.

2.3 Arbitrage Error

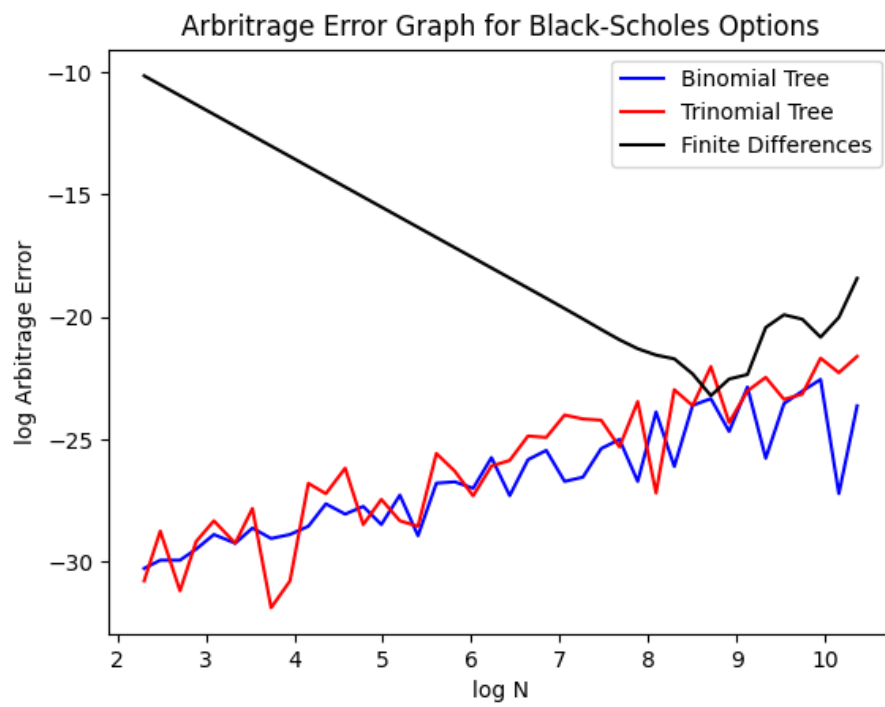


Figure 3: Effect of Steps on Arbitrage Error

2.4 Runtime

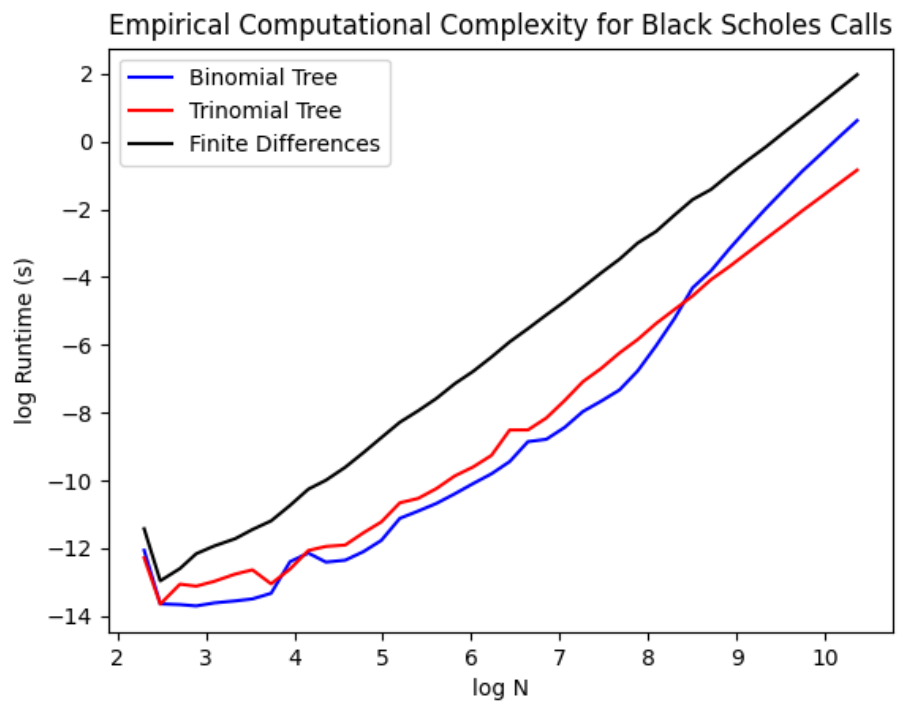


Figure 4: Effect of Steps on Arbitrage Error

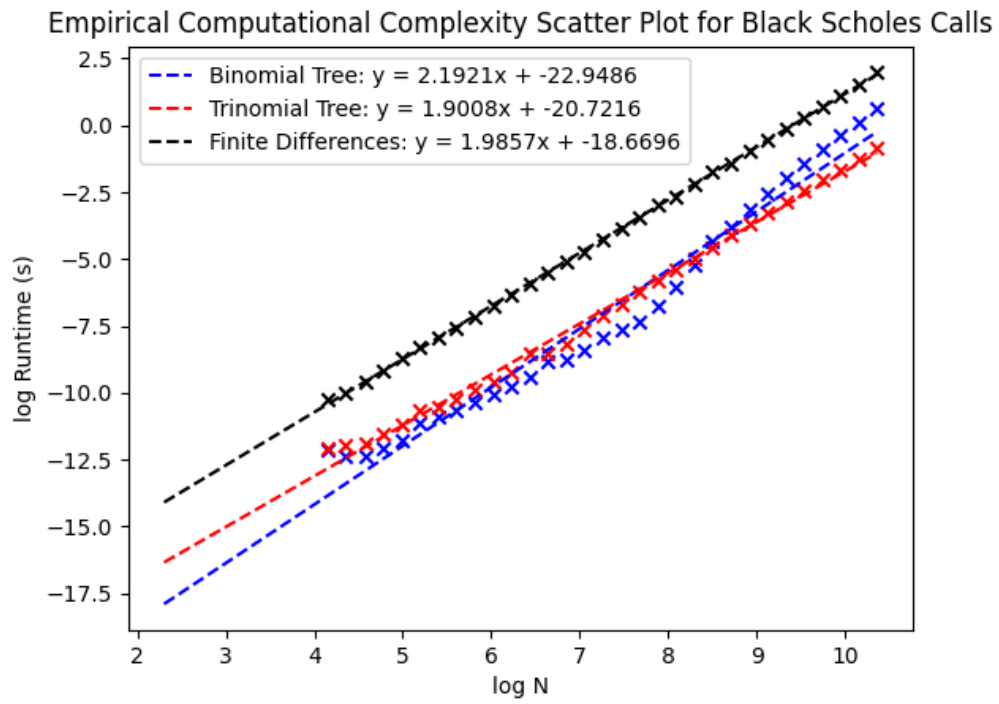


Figure 5: Effect of Steps on Arbitrage Error

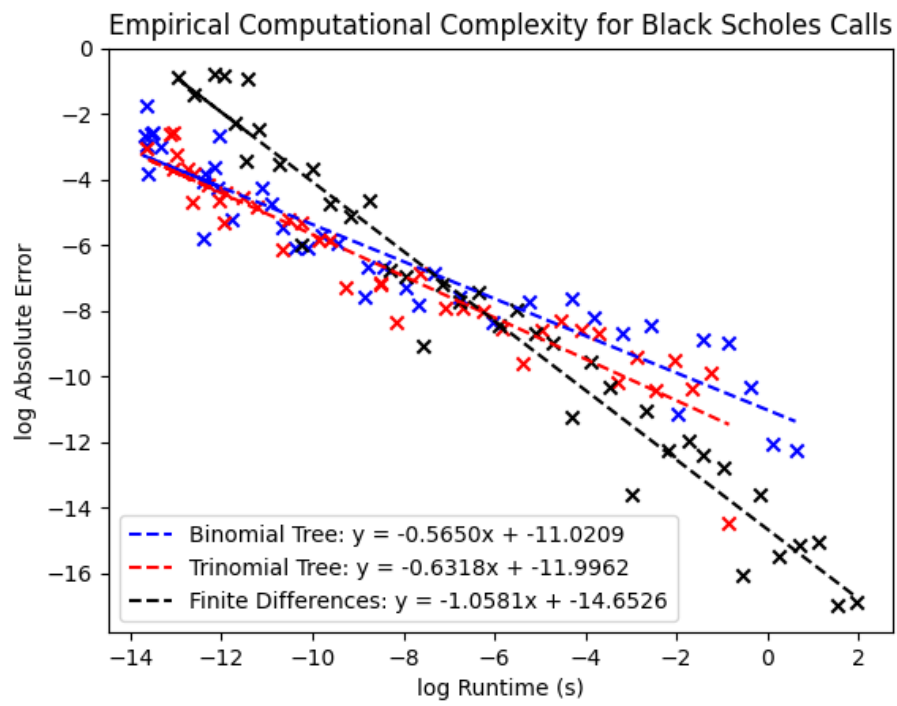


Figure 6: Effect of Steps on Arbitrage Error

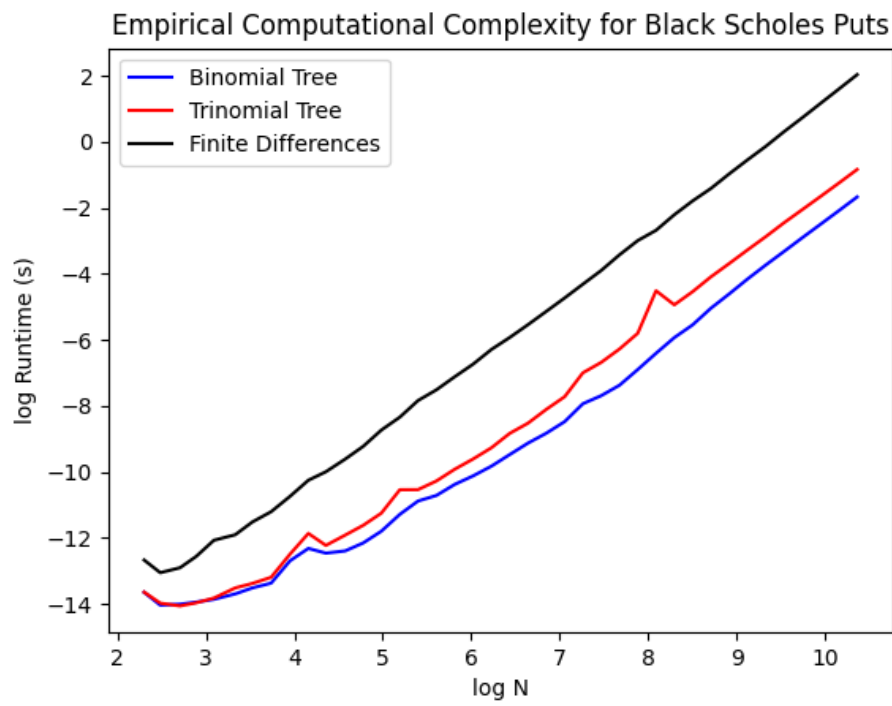


Figure 7: Effect of Steps on Arbitrage Error

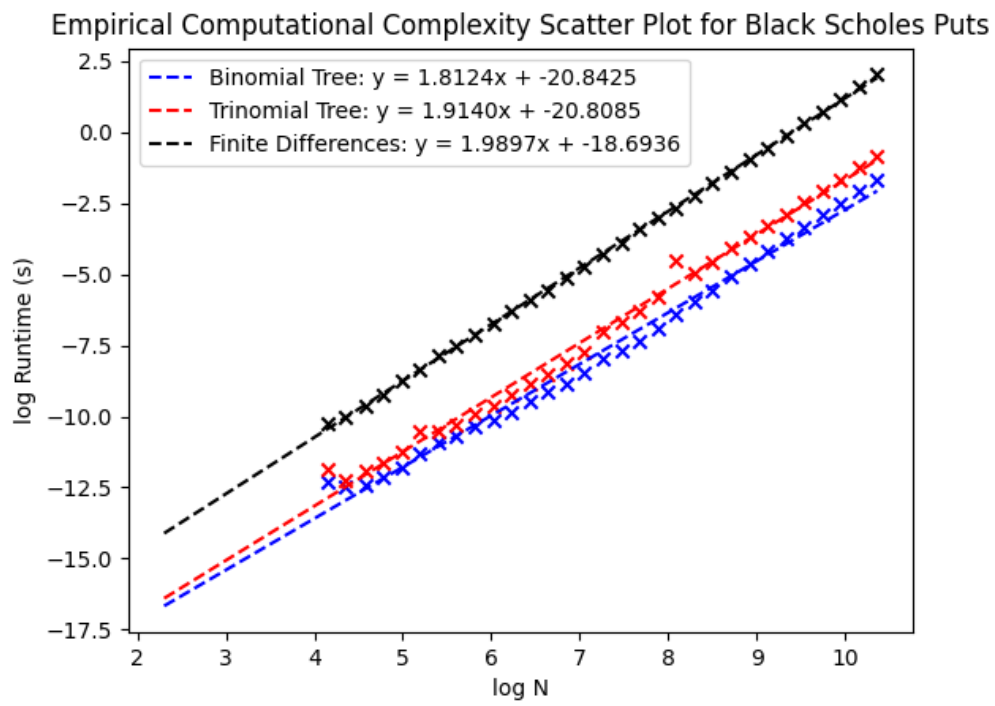


Figure 8: Effect of Steps on Arbitrage Error

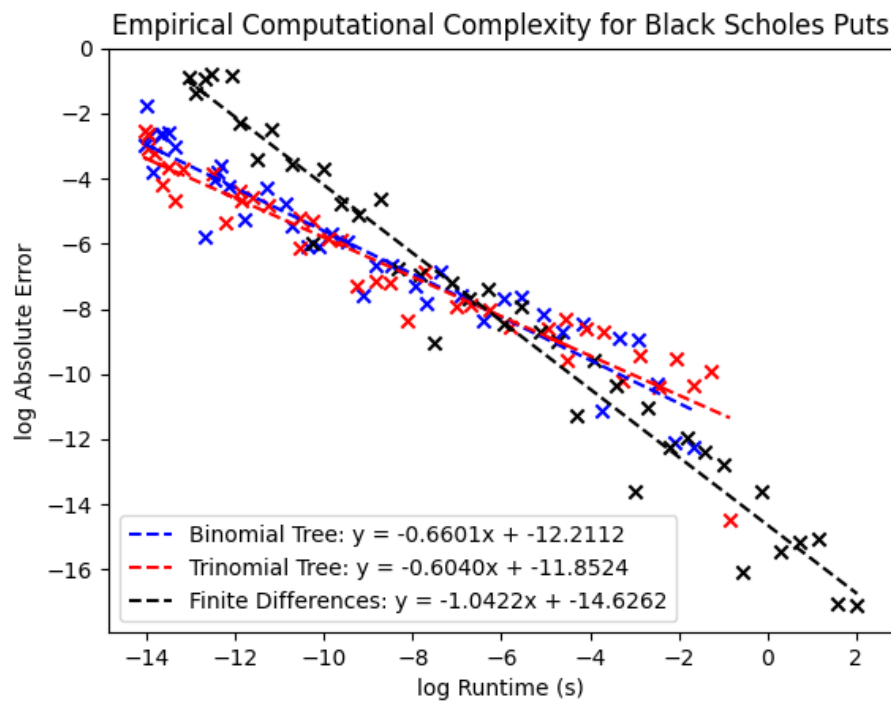


Figure 9: Effect of Steps on Arbitrage Error

2.5 Memory

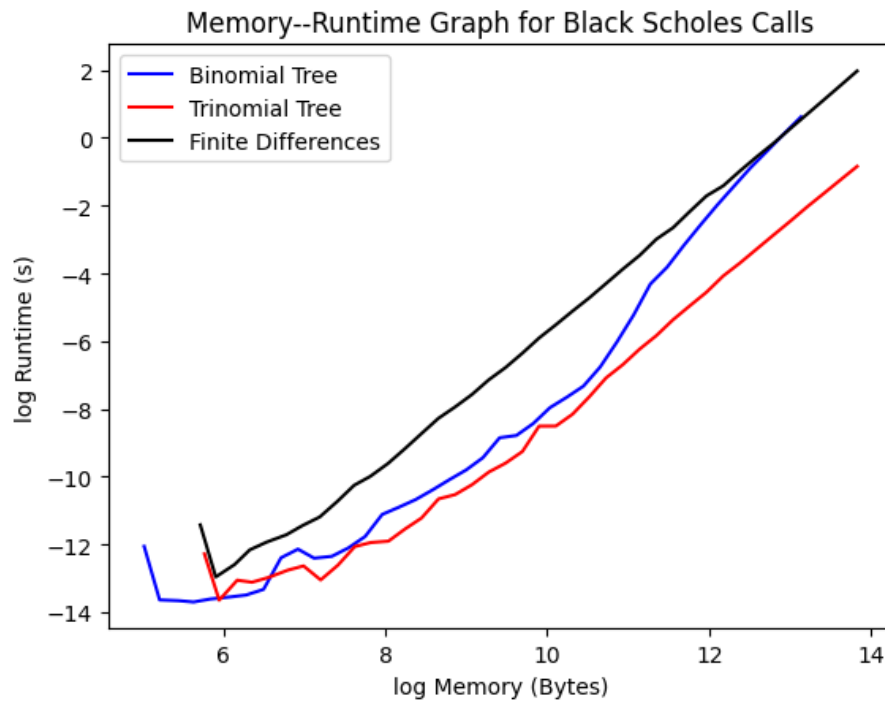


Figure 10: Memory to Runtime Relationship for Call Options